Convex Optimization Term Project - Proposal

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Topic and Motivation:

This project will study optimization based path planning for a 2D point robot that is navigating a specific obstacle course. We hope to understand how the different objective (cost) functions and optimization methods affect path quality, convergence behavior, and robustness. This study on a 2D point robot provides a representative model to evaluate different gradient-based and/or constrained optimization methods and cost function definitions for robotics path planning.

Scope:

Our group will implement several different optimization methods, including both gradient descent and Newton's method. Both are well documented baselines for potential field navigation. Potential fields are known to have local minima based on the configuration of the obstacles in the environment (i.e. tight passageways). To explore this limitation, we will implement gradient descent with momentum, which should allow the robot to escape local minima. Since these methods are all gradient-based, we will also implement augmented Lagrangian and quasi-Newton optimization for comparison. We will also study the effect of the cost function by varying the repulsive forces exhibited by objects and the attractive force exhibited by the goal.

Technical Plan and Success Criteria:

The environment will consist of 2D maps with fixed start and goal points and static obstacles. The paths will be represented as discrete waypoints with the interior points treated as the optimization variables. We plan on having 3 different types of testing environments (easy, medium, and hard pathing difficulty) in total. A cost function will be constructed based on each testing environment, with repulsive forces from each object and an attractive force from the goal. The optimization methods to be tested will be a standard gradient descent, Newton's method, gradient descent with momentum, augmented Lagrangian, and quasi-Newton. To evaluate the results, we will use success rate (does robot reach the goal) and iteration count as the metrics.

The project will be considered successful if the following criteria are met: We can correctly implement multiple objective formulations and optimization methods. The system produces valid collision free pathing across the different environments. Evaluation shows clear differences in performance. Lastly, analysis explains observed behaviors such as convergence speed or local minima.

Team Roles:

Member 1 (Nihal): Investigate gradient-descent methods and build simulation/map environments.

Member 2 (Adam): Investigate Newton and quasi-Newton methods and potential field formulations.

Member 3 (Vineal): Investigate Augmented Lagrangian methods and form final experiment pipeline/collect results.

Each member will be responsible for implementing their own optimization method, which has fundamentally different problem structures (constrained vs unconstrained approaches). Reducing to two methods would weaken the study to two methods that differ minimally (i.e step-size). Instead of splitting different aspects of each method among two members, each member will be able to fully understand their optimizer pipeline along with a specific part of the project. By splitting the project setup into three chunks, we reduce any gaps of knowledge for the project as a whole, and allow each member to focus on implementing an optimization method and their relevant part of the project setup.

Relevant Technical References:

Real-Time Obstacle Avoidance for Robotic Manipulators - this paper introduced the concept of potential fields, how to model objects using them, and how to use partial derivatives of the potential fields to influence robot control.
https://khatib.stanford.edu/publications/pdfs/Khatib_1986_IJRR.pdf

Modified Newton's Method Applied to Potential Field-Based Navigation for Mobile Robots - this paper studies Newton's method to decrease the oscillation in traditional gradient descent methods when using potential fields.
https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=1618532

Artificial Potential Field Path Planning - this Github repo provides code to run and visualize robot navigation using potential fields with both static and dynamic obstacles. We hope to replicate similar styles of experiments for our project.
https://github.com/snktshrma/artificial-potential-field-path-planning